

Modulating temperature and pH during subcritical water extraction tunes the molecular properties of apple pomace pectin as food gels and emulsifiers

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ABSTRACT

This study explores the cascade extraction of polyphenols and pectins from apple pomace and their potential application as food emulsifiers. The extraction of polyphenols was achieved by a simple method using ethanol and resulted in remarkable antioxidant activity. Pectic polysaccharides were sequentially extracted (5, 10 and 15 min) using subcritical water at two different temperatures (120 °C and 140 °C) and three different pH conditions (pH 3.0, 5.0 and 7.0). Acidic pH resulted in higher pectin yields when combined with higher temperature, whereas neutral conditions were less selective towards pectin. Pectin with different branching patterns could be isolated by tuning the process conditions. Linear homogalacturonan (HG) was preferentially isolated at milder conditions, whereas more branched rhamnogalacturonan (RG-I) populations required longer extraction times. Subcritical water enabled the extraction of pectins with different degrees of methoxylation (DM) depending on the extraction pH, which enabled the production of pectin acid gels with varying rheological properties. Low DM pectin was directly recovered from apple pomace at lower pH, eliminating the need for its conventional derivatization from high DM pectin. The emulsifying properties of two apple pomace pectin fractions with high and low methoxyl content were compared; the pectin with higher methoxyl content showed higher emulsifying capacity and stability. This study demonstrates that the relative abundance of the different pectic domains and the methoxylation of pectin can be controlled by subcritical water extraction, which has important implications for the valorization of pectic-rich agricultural wastes into valuable food ingredients.

1. Introduction

The food industry generates large quantities of residual streams after harvesting and processing, with a significant economic, environmental and social impact. As an example, worldwide apple production has reached 80 million tons annually (FAOSTAT, 2018), providing a vast amount of input to the production of apple juice, cider, syrup, and vinegar. Industrial processing of apple generates a useful by-product, apple pomace, that has been traditionally used as animal feed. However, the recovery rate of apple pomace into feed is generally very low (Lyu, et al., 2020), therefore it is either used as biogas or discarded

directly to the soil (Shalini & Gupta, 2010). Although research efforts have been made for efficient waste management for apple pomace (Shalini et al., 2010), smarter and cost-effective strategies are required for its valorization in a biorefinery concept, as it constitutes a rich source of phenolic compounds and dietary fibres.

Apple pomace is composed of skin and flesh (95%), seeds (2–4%) and stems of apple (1%); with a content of pectin (3.5–14.3%), cellulose (13%), hemicelluloses (4.3%), starch (5.6%), fat (0.6–4.2%), protein (1.1–4.7%), polyphenols, vitamins and minerals (Lyu, et al., 2020; Perussello, Zhang, Marzocchella, & Tiwari, 2017). The main polyphenol groups present in apple pomace are phenolic acids (*p*-coumaric, caffeic,

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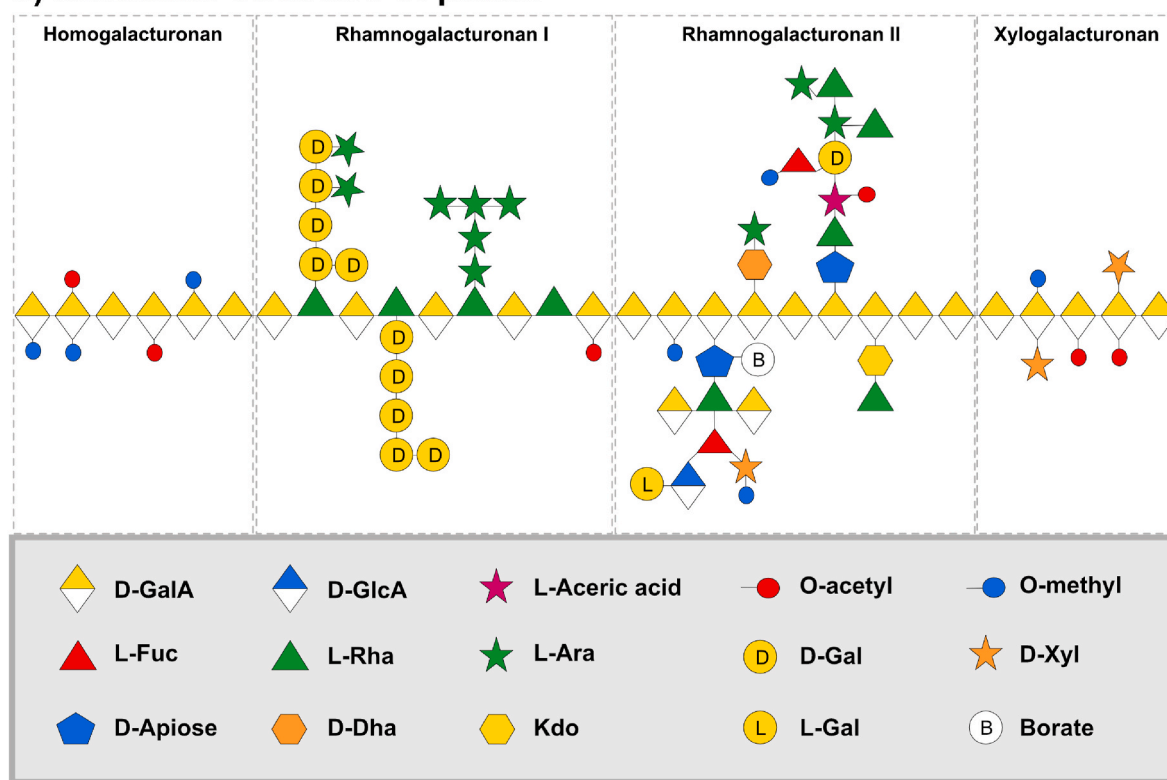
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chlorogenic, protocatechuic), flavanols (epicatechin, procyanidin B2), dihydrochalcones and flavonols (hyperin, quercetin, avicularin) (Díneiro García, Valles, & Picinelli Lobo, 2009; Schieber et al., 2003). The rich content of polyphenols in apple pomace renders it a valuable source to be used as a potential food additive conferring oxidative stability. As natural antioxidant compounds, polyphenols derived from plant sources can be incorporated into food and drinks to increase shelf life by preventing lipid and protein oxidation as well as protecting against microbial activity (Bouarab Chibane, Degraeve, Ferhout, Bouajila, & Oulahal, 2019; Falowo, Fayemi, & Muchenje, 2014; Pasrija, Ezhilarasi, Indrani, & Anandharamakrishnan, 2015). Furthermore, plant polyphenols hold potential to be used in cosmetic formulations thus replacing the use of synthetic antioxidants (Gasparrini, et al., 2017; Saewan & Jimtaisong, 2015; Zillich, Schweiggert-Weisz, Eisner, & Kerscher, 2015).

As another large constituent of apple pomace, pectin is a structurally heterogeneous polysaccharide typically composed of four domains with

distinct branching patterns, namely homogalacturonan (HGA), rhamnogalacturonan-I (RG-I), rhamnogalacturonan-II (RG-II), and xylogalacturonan (XGA) (Fig. 1a). The main sugar constituent of the backbone of the different pectin domains is (1 → 4)-linked- α -D-galacturonic acid (GalA). HGA is defined as the smooth region of pectin, with acetyl and methyl esterification. XGA consists of an HGA backbone substituted by β -D-xylopyranose (Xyl), and methyl and acetyl esters. RG-I is a heteropolymer of alternating (1 → 2)-linked- α -L-rhamnopyranose (Rha) and (1 → 4)-linked- α -D-galacturonic acid units with substitutions of α -D-galactopyranose (D-Gal) and α -L-arabinofuranose (Ara). RG-II is known as the hairy region of pectin and contains the richest diversity of glycosidic linkages and sugars including L-Ara, L-Rha, D/L-Gal, D-glucuronic acid (GlcA), apiose, aceric acid (3-C'-carboxy-5-deoxy-L-xylose), 2-O-methyl-L-fucose, 2-O-methyl-D-xylose, Kdo (3-deoxy-D-manno-2-octulosonic acid) and Dha (3-deoxy-D-lyxo-2-heptulosaric acid). The smooth regions of pectin molecules can form a network via intermolecular bonds, resulting in gelation (Kastner, et al., 2014), depending on

a) Molecular structure of pectin



b) Sequential extraction of phenolics and pectin from apple pomace

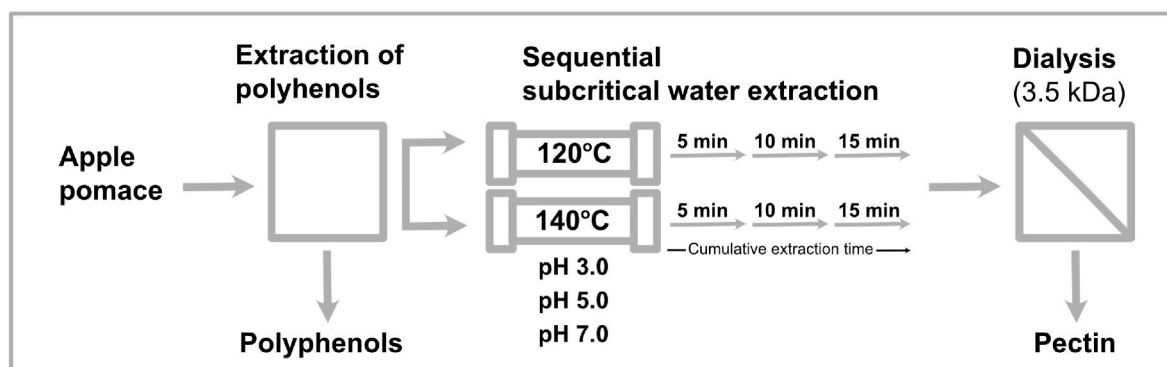


Fig. 1. (a) Molecular structure of pectin in apple pomace, (b) Sequential extraction process of polyphenols and pectin from apple pomace.

the degree of methylation (DM). The gelation of high methoxyl pectin (HMP) (DM>50%) is driven by hydrophobic interactions and hydrogen bonds, and occurs in the presence of a co-solute (typically simple sugars) in acidic pH (≤ 3.5) (Tsoga, Richardson, & Morris, 2004). Low methoxyl pectin (LMP) (DM<50%), on the other hand, requires divalent cations (typically Ca^{2+}) for gelation, which forms intermolecular ionic junction zones between dissociated carboxyl groups. Due to this gelation capacity, pectin finds wide uses in the food industry as a gelling and thickening agent. LMP is particularly of interest as viscosifier to produce low-sugar products, specifically targeted for consumers with diabetes or obesity (Muñoz-Almagro, Montilla, & Villamiel, 2021). In the food industry, LMP is typically derived from HMP that is extracted from commercial pectin sources (e.g. citrus peel) (May, 1990). This requires the implementation of an additional processing step, increasing production costs. Direct extraction of LMP from plant tissues would, therefore, eliminate these additional steps. Conventional extraction of pectin is carried out using acid solutions and high temperature, despite its water-soluble characteristics. The use of relatively harsh conditions are required to disrupt supramolecular interactions between pectin and other plant cell wall components, which are known to be resistant to extraction (Sista Kameshwar & Qin, 2018). Subcritical water extraction (SWE) constitutes a promising technique for the extraction of matrix polysaccharides from recalcitrant plant cell wall tissues such as wood and grasses (P. O. Kilpeläinen et al., 2014; Ruthes et al., 2020). SWE uses water in liquid state at high pressure as a solvent, which undergoes physical changes such as lower viscosity and polarity, facilitating the extraction and diffusion of the target biomolecules (Plaza & Turner, 2015). In previous works, we have demonstrated that controlling the pH using specific buffers under SWE is an efficient tool to preserve the native functionalities (i.e. acetylation and feruloylation) and to tune the physicochemical characteristics of polysaccharides from plant biomass using a single setup (Rudjito, Jiménez-Quero, Hamzaoui, Kohnen, & Vilaplana, 2020; Rudjito, Ruthes, Jiménez-Quero, & Vilaplana, 2019; Ruthes, Martínez-Abad, Tan, Bulone, & Vilaplana, 2017).

In this work we hypothesized that the combination of acidic pH under subcritical water conditions would enable the extraction of pectins from apple pomace residues with a controlled range of DMs. We propose for the first time a cascade extraction process for the sequential recovery of polyphenols and pectin from apple pomace. We evaluated the antioxidant activity of the extracted polyphenols, and we monitored the time evolution of the pectin extraction at two different temperatures (120 °C and 140 °C) and 3 different pH conditions (pH 3.0, 5.0 and 7.0). The influence of the extraction conditions on the sugar composition, molar mass distribution, degree of methylation and rheological properties of the isolated pectins was also analyzed. Finally, as a proof of concept we evaluated the use of the apple pomace pectins extracted by SWE as vegetable oil emulsifiers together with their long-term stability.

2. Materials and methods

2.1. Materials

Apple pomace was kindly provided by Vaddö Musteri (Grisslehamn, Sweden) as a pressed fruit cake in wet state. All chemicals, standards and reagents were purchased from Sigma-Aldrich (Sweden). Spectra/Por 3 (3.5 kDa MWCO) dialysis membranes were purchased from SpectrumLabs (Repligen, USA). Total starch assay kit was purchased from Megazyme (Wicklow, Ireland) and the dye reagent for the Bradford protein assay was from Bio-Rad Laboratories (Sweden).

2.2. Extraction processes

2.2.1. Extraction of polyphenols

The proposed cascade extraction of polyphenols and pectins from apple pomace is presented in Fig. 1b. For the extraction of polyphenols, apple pomace was milled in wet state using a kitchen blender and

suspended in 50% (v/v) ethanol at 1:10 (pomace:solvent) ratio, following a procedure previously developed for sunflower hulls (Carolin Menzel, González-Martínez, Chiralt, & Vilaplana, 2019). The suspension was stirred for 1 h at room temperature in the dark. Ethanol was evaporated using a rotary evaporator (Rotavapor R-210, Büchi Labor-technik AG, Switzerland) at 36 °C under vacuum in the dark and the concentrated phenolic extract was freeze-dried.

2.2.2. Extraction of pectins

Subcritical water extraction (SWE) of polysaccharides from apple pomace was performed using an accelerated solvent extractor system (Dionex, ASE™ 350, Thermo Scientific™ Sunnyvale, CA, USA), following a similar approach previously developed for cereal bran (Ruthes, et al., 2017). The remaining apple pomace after the extraction of polyphenols was freeze-dried, weighed to 2 mg and then mixed with inert material in the extractor cells. Three cycles of sequential extractions were performed at pH 3.0, 5.0 and 7.0 for 5, 10 and 15 min at 120 °C and 140 °C. The extraction solvents were buffered with 0.1 M citrate buffer. The extracts were dialyzed using 3.5 kDa MWCO membranes for 72h and then freeze-dried. Extractions were performed in duplicate for each pH and temperature condition.

2.3. Analytical methods

2.3.1. Starch and protein content

The starch content of apple pomace was determined in triplicate using the Megazyme total starch assay kit (Megazyme, Wicklow, Ireland). The soluble protein content of apple pomace was determined by the dye-binding Bradford assay (Bradford, 1976). 100 µL dilutions of apple pomace (10 mg/mL) were mixed with 900 µL of diluted (5x) dye reagent and incubated at room temperature for 15 min. The absorbance of the samples was measured at 595 nm against deionized water as blank. Soluble protein content was calculated (in triplicate) using the standard calibration of bovine serum albumin between 0.025 and 0.2 mg/mL.

2.3.2. Total phenolic content

Total phenolic content (TPC) of the extracted polyphenols was determined using the Folin-Ciocalteu assay. 100 mg/mL solutions of the samples were prepared in 50% ethanol and then filtered on filter paper. 100 µL aliquots of 10x dilutions of the solutions (10 mg/mL) were mixed with 100 µL of Folin-Ciocalteu reagent (2 N). After 2 min, 800 µL of sodium carbonate solution (20%, w/v) was added and the mixtures were incubated at 40 °C for 20 min in the dark. The absorbance was measured spectrophotometrically at 765 nm using a UV-Vis spectrophotometer (Varian Cary 50 UV-VIS, Agilent) against water as blank. Gallic acid solution (0.01–0.08 mg/mL) was used as standard and the TPC of the extracted polyphenols was expressed in gallic acid milliequivalents (mg GAE)/g apple pomace. The measurements were performed in duplicates.

2.3.3. Evaluation of antioxidant activity

The radical scavenging activity of the extracted polyphenols was determined by the 1,1-diphenyl-2-picrylhydrazyl (DPPH•) assay in duplicate according to Brand-Williams et al. (Brand-Williams, Cuvelier, and Berset (1995) with minor modifications. Briefly, 10 mg/mL solutions (w/v methanol) of the phenolic extracts were mixed with 0.3 mM methanolic DPPH• solution and incubated for 30 min at room temperature in the dark. The absorbances of the reaction media were measured at 517 nm using a microplate reader (Clariostar Plus, BMG LABTECH). The radical scavenging activity of the samples was calculated based on the decrease in the absorbance of DPPH• and presented as half-maximal effective concentration (EC₅₀) (mg extract/mg DPPH•).

2.3.4. Phenolic acid profiling

The phenolic acid content of the extracted polyphenols was

determined in duplicate using a Waters HPLC system (Waters 2695 separation module, Waters 2996 photodiode array detector, USA) coupled to a UV/Vis detector, equipped with a C18 guard column and an SB-C18 separation column (Zorbax SB-C18 5 μm particle size, 4.6 $\times$ 250 mm, Agilent, Santa Clara, CA, USA). Before the freeze-drying step, the fresh polyphenolic extract was filtered (0.2 μm nylon filter) and then diluted 100 times using water and injected into HPLC. A gradient of methanol and 2% acetic acid was used as the mobile phase at a flow rate of 1 mL/min at 25 °C as described in C. Menzel, Gonzalez-Martinez, Chiralt, and Vilaplana (2019). The standard calibration was performed using caffeic acid, chlorogenic acid, *p*-coumaric acid and ferulic acid (0.005–0.1 g/L) at 325 nm.

2.3.5. Determination of extraction yields

Yields of SWE at each condition were determined gravimetrically and calculated on dry basis. The moisture content of apple pomace was determined gravimetrically after freeze-drying.

2.3.6. Monosaccharide composition

The monosaccharide composition of the apple pomace, SW extracts and residues was analyzed by high-performance anion-exchange chromatography with pulsed amperometric detection (HPAEC-PAD) after methanolysis and acid hydrolysis, respectively. On one hand, the apple pomace and hydrothermal (SWE) extracts were hydrolyzed in triplicate by 2-step methanolysis using 1–2 mg samples (Martínez-Abad, Giummarella, Lawoko, and Vilaplana (2018). Secondly, the apple pomace and residues after SWE were treated by 2-step sulfuric acid hydrolysis in triplicate (Saeman, Moore, Mitchell, & Millett, 1954). 3–4 mg sample was mixed with 72% sulfuric acid and incubated at room temperature for 1 h followed by dilution and hydrolysis at 100 °C for 3 h. The samples were then filtered with 0.45 μm nylon filters and diluted with ultrapure water (10x).

The hydrolyzed samples were injected into the HPAEC-PAD system (Dionex ICS 3000, Sunnyvale, USA) equipped with a CarboPac PA1 column. Two gradient programs were used for the separation of neutral and uronic sugars, respectively (McKee, et al., 2016) and their quantification was performed by external standard calibration (0.005–0.1 g/L): fucose (Fuc), arabinose (Ara), rhamnose (Rha), galactose (Gal), glucose (Glc), xylose (Xyl), mannose (Man), galacturonic acid (GalA), glucuronic acid (GlcA) and 4-O-methyl glucuronic acid (meGlcA).

2.3.7. Molar mass distribution

The molar mass distribution of the hydrothermal extracts was determined by size exclusion chromatography (SEC). The samples were dissolved in the SEC eluent consisting of DMSO supplemented with 0.5% (w/w) lithium bromide (LiBr) to give a concentration of 3 mg/mL. The samples were then injected into SEC (SECcurity 1260, Polymer Standard Services, Mainz, Germany) with a refractive index detector using the conditions reported by Ruthes et al. (2017). A standard calibration was performed using pullulan standards (342–708 000 Da).

2.3.8. Fourier transform infrared spectrometry (FTIR)

Infrared spectra of the selected extracts (120 °C-pH 3.0–10 min, 120 °C-pH 5.0–10 min, 120 °C-pH 7.0–10 min, 140 °C-pH 3.0–5 min, 140 °C-pH 3.0–10 min, 140 °C-pH 3.0–15 min) were collected using a PerkinElmer Spectrum 100 FTIR instrument (Norwalk, CT) equipped with a single reflection accessory unit (Golden Gate, Graseby Spec Ltd, Kent, England). FTIR data were recorded after 16 scans for each sample at a resolution of 4.0 cm^{-1} for a spectral range from 4000 to 600 cm^{-1} . The spectra of the samples were automatically smoothed and corrected by the Spectrum software (PerkinElmer).

2.3.9. Degree of methylation

The degree of methylation (DM) of the extracted pectins was determined according to Klavons et al. (Klavons and Bennett (1986). The samples were saponified at 20 °C overnight using 2 M NaOH and then

neutralized by 12 M HCl solution. Oxidation of methyl esters to form-aldehyde was carried out using alcohol oxidase solution from *Pichia Pastoris* (884.5 U) at 25 °C for 15 min. The samples were then complexed to diacetyldihydrolutidine by mixing with 0.02 M acetylacetone in 2 M ammonium acetate and 0.05 M acetic acid at 58 °C for 15 min. The absorbance of the resulting samples was measured at 412 nm against diluted alcohol oxidase solution as blank. The standard calibration was performed using methanol at 0.01–1.00 mg/mL. The measurements were performed in duplicate.

2.4. Gelling properties of the pectin fractions

2.4.1. Preparation of pectin gels

To evaluate pectin gelation, the selected hydrothermal extracts were dissolved in citrate buffer at pH 3.5 to give a pectin concentration of 5% (w/v). 60% (w/v) sucrose was added to the extracts and the mixtures were heated at 100 °C for 10 min for complete solubilization. The samples were stored at 4 °C until further analysis.

2.4.2. Rheological properties

The rheological properties of the pectin gels were measured using a strain-controlled rheometer (ARES G2, TA Instruments, New Castle, DE, USA) at 25 °C with a 20 mm diameter plate as upper geometry, and a gap of 50 μm . Viscosity measurements were performed in a shear rate range between 1 and 100 s^{-1} . Viscoelasticity tests were performed by frequency sweep test with 0.5% oscillation strain, selected from the linear viscoelastic region, and between 1 and 125 rad s^{-1} .

2.5. Emulsifying properties of the pectic fractions

The emulsifying properties of two selected extracts from apple pomace with high and low methoxyl content were benchmarked with a commercially available high methoxyl citrus pectin control (DM of 71%) gently provided by Ceamsa (Porriño, PO, Spain) (Roman, et al., 2021).

2.5.1. Preparation of emulsions

Oil-in-water emulsions were prepared according to the procedure employed by Alba (Alba, Dimopoulou, & Kontogiorgos, 2021) and Cuevas-Bernardino (Cuevas-Bernardino, Lobato-Calleros, Román-Guerrero, Alvarez-Ramirez, & Vernon-Carter, 2016) with some modifications. The aqueous phase of the emulsions were prepared by hydrating the corresponding pectin at 1.1% w/v in 0.01M hydrochloric acid-potassium chloride (HCL/KCL) buffer (pH = 2). Pectin solutions were hydrated under stirring at room temperature until complete hydration. Pre-emulsions were prepared by adding 2 mL of refined sunflower oil to 18 mL of the pectin solution in order to yield a volume fraction $\phi = 0.1$ and nominal pectin concentration of 1% v/v. The stock emulsion was fabricated at room temperature by high speed homogenization using an high shear Ultra-Turrax T25 homogenizer (IKA, USA) at a rotational speed of 10,000 rpm during 3 min. Subsequently, an ultrasonication step was applied using a UP200S sonicator (Hielscher Ultrasonics GmbH, Germany) operating at 24 KHz and equipped with a Microtip S3 probe. Ultrasonication was applied for 1 min with pulse ultrasounds (50% per second) at a 100% amplitude. Droplet size distribution was measured immediately after the emulsion preparation (fresh). Emulsions were kept at 4 °C in order to evaluate their long-term stability up to 10 days (3, 5, 8 and 10 days).

2.5.2. Droplet size distribution in emulsions and long term stability

Droplet size was measured immediately after making the emulsion. In order to determine the long-term stability of the emulsion, droplet size was measured during 10 days (2, 3, 5, 8, 10 days). The average volume weighted diameter $d(3,4)$ of emulsion droplets was determined in triplicate using a Masterizer 2000 (Malvern Instruments Ltd., Malvern, UK). Refractive index of pectins and dispersion medium were established at 1.464 and 1.33, respectively.

3. Results and discussion

3.1. Composition of the starting apple pomace

As a by-product of apple processing, apple pomace was evaluated in this study as a rich source of phenolic acids and dietary fibre pectin to be valorized into antioxidant gels and emulsifiers. The composition of the initial raw pomace was characterized in terms of moisture, protein and carbohydrate content (Table 1). The apple pomace contained 80.9% moisture as in agreement with previous studies (Fernandes, Ferreira, et al., 2019). Soluble proteins comprised a small portion (1.2%) of apple pomace and carbohydrates constituted its major components. As pectins, cellulose and hemicelluloses are the main polysaccharides of apple pomace (Perussello, et al., 2017), both sulfuric acid hydrolysis and two-step methanolysis were performed for the estimation of cellulose and pectin/hemicelluloses (neutral and uronic sugars), respectively. Sulfuric acid hydrolysis showed that fructose was the major monosaccharide in the apple pomace, followed by glucose. This glucose was attributed to cellulose, hemicellulose (e.g. xyloglucan and β -glucan) (Fernandes, Silva, et al., 2019) and starch (6.8%). Methanolysis, on the other hand, revealed the high abundance of pectic monosaccharides (GalA, Xyl, Fuc, Ara, Gal, Rha and GlcA), indicating the occurrence of different pectin regions (RG-I, RG-II, XGA and HGA). No fructose was detected after methanolysis, which was attributed to its complete degradation at hydrolysis conditions by furanoid dehydration (Becker, et al., 2021).

3.2. Characterization of extracted polyphenols

The phenolic content and radical scavenging activity of the ethanolic extract were presented in Table 2. The ethanolic extract yielded a TPC of 8.2 mg GAE/g apple pomace, in agreement with previous studies where similar extraction methods were used (Ajila, Brar, Verma, Tyagi, & Valéro, 2011; Chandrasekar, Martín-González, Hirst, & Ballard, 2015). The content, as well as the profile of phenolics in apple pomace, varies depending on cultivar, the maturity of apple, growing and storage conditions and parts of apple that apple pomace was produced from (Diñeiro García, et al., 2009; Kalinowska, Bielawska, Lewandowska-Siwkiewicz, Priebe, & Lewandowski, 2014). The phenolic acid profile of the freshly extracted polyphenols, as analyzed by HPLC, showed a dominating content of chlorogenic acid (Supplementary Fig. S1) in agreement with previous studies (Diñeiro García, et al., 2009; Schieber et al., 2003). Five unknown compounds were observed between 32 and 37 min of the chromatogram (Supplementary Fig. S1), which may represent various phenolic families such as flavonols or dihydrochalcones (Diñeiro García, et al., 2009). However, these were

Table 2

Characterization of phenolics extracted from apple pomace.

	Freeze-dried polyphenol extract ^c
Total phenolic content (mg GAE/g) ^a	8.2 $\pm$ 0.0
Chlorogenic acid content (μ g/g) ^b	35.5 $\pm$ 0.5
EC ₅₀ (mg/mg DPPH)	19.4 $\pm$ 0.9

^a Total phenolic content was determined using Folin-Ciocalteu reagent.

^b Chlorogenic acid content was determined by HPLC analysis.

^c The results are presented as averages $\pm$ standard deviation.

not assigned to any phenolic compound as the UV-spectra did not correspond to the available standards. As for the radical scavenging activity of the extracted polyphenols, 19.4 mg extract was necessary to reduce 50% of 1 mg of DPPH[•], suggesting a strong antioxidant activity that was attributed to the high phenolic content of the extract.

3.3. Subcritical water extraction of apple pomace pectins

The influence of extraction parameters (temperature, pH and time) under subcritical water conditions on extraction yields, monosaccharide composition and molar mass distributions of pectins from apple pomace was evaluated. As expected, a higher extraction temperature of 140 °C promoted higher extraction yields compared to extraction at 120 °C (Fig. 2a); this increase in yields at higher temperatures was observed at all tested pH conditions (pH 3, 5, and 7) but it was more significant at pH 7. Interestingly, when evaluating the evolution of the extraction yields with time, the highest yields at 140 °C were observed in the first cycle (5 min), whereas longer times were required at 120 °C to achieve similar yields (Fig. 2b). This confirms that increased temperature promotes mass transfer under subcritical water conditions, resulting in improved extraction kinetics in terms of extracted polysaccharides in shorter times.

At 120 °C, the total solid yields of SWE at all pH values were similar to each other (Fig. 2b), indicating that different pH conditions do not have a substantial effect on the overall extraction yields at this temperature. However, at 140 °C extraction at neutral conditions (pH 7) resulted in higher overall yields compared to more acidic conditions (pH 5 and pH 3), which may indicate on one hand the preferential extraction of specific polysaccharide components at such neutral conditions, and the occurrence of degradation of labile polysaccharide moieties at more acidic conditions. Interestingly, at 140 °C, a gradual drop was observed in the total yields at all pH conditions as the extraction time extended, differently from the SWE at 120 °C. This could be due to the fact that the available pectins were already accessed at the earlier stage of the extractions at this higher temperature. Our results agree with the optimal extraction of pectic polysaccharides at 140 °C, preventing extensive pectic degradation at higher temperatures (Benito-Román, Alonso-Riaño, Díaz de Cerio, Sanz, & Beltrán, 2022; Zhang et al., 2022).

3.3.1. Time evolution of the monosaccharide composition during SWE

The influence of process conditions (temperature, pH and time) on the monosaccharide composition of the extracted fractions under subcritical water was evaluated (Fig. 2c, Table 3). In general, all the extracts showed a predominant content of monosaccharides present in pectins such as GalA, Ara, Gal and Rha, indicating the selective extraction of pectic polysaccharides, mainly linear HG and RG-I. The low presence of Fuc and GlcA in all the extracts at different time and pH conditions may indicate that the presence of RG-II was limited. In parallel, the residues exhibited higher content in Glc, Xyl and Man, which indicates that cellulose and hemicelluloses (xylans and mannans) were enriched in the residue and not extracted at the tested SWE conditions. In previous works, higher temperatures (160 °C and 170 °C) have been applied to isolate polymeric hemicelluloses from different cereal and wood biomasses (P. Kilpeläinen et al., 2012; Martínez-Abad, et al., 2018; Rissanen et al., 2014; Rudjito et al., 2020; Ruthes et al., 2017). This

Table 1

Chemical composition of apple pomace.

Moisture (%)	80.9 $\pm$ 0.2	
Soluble protein (%)	1.2 $\pm$ 0.0	
Total phenolic (mg GAE/g DW)	8.2 $\pm$ 0.0	
Starch (%)	6.8 $\pm$ 1.1	
Monosaccharide analysis	Acid hydrolysis	Methanolysis
Total Carbohydrate (mg/g DW)	879.7 $\pm$ 117.3	443.5 $\pm$ 6.4
Ara (%)	3.3	13.4
Fuc (%)	4.6	1.8
Fru (%)	59.0	n.d.
Gal (%)	2.2	5.9
Glc (%)	26.8	52.2
Man (%)	0.4	1.3
Rha (%)	n.d.	2.0
Xyl (%)	2.0	5.1
GalA (%)	1.7	17.3
GlcA (%)	n.d.	1.0

NOTE. The results are presented as averages $\pm$ standard deviation.

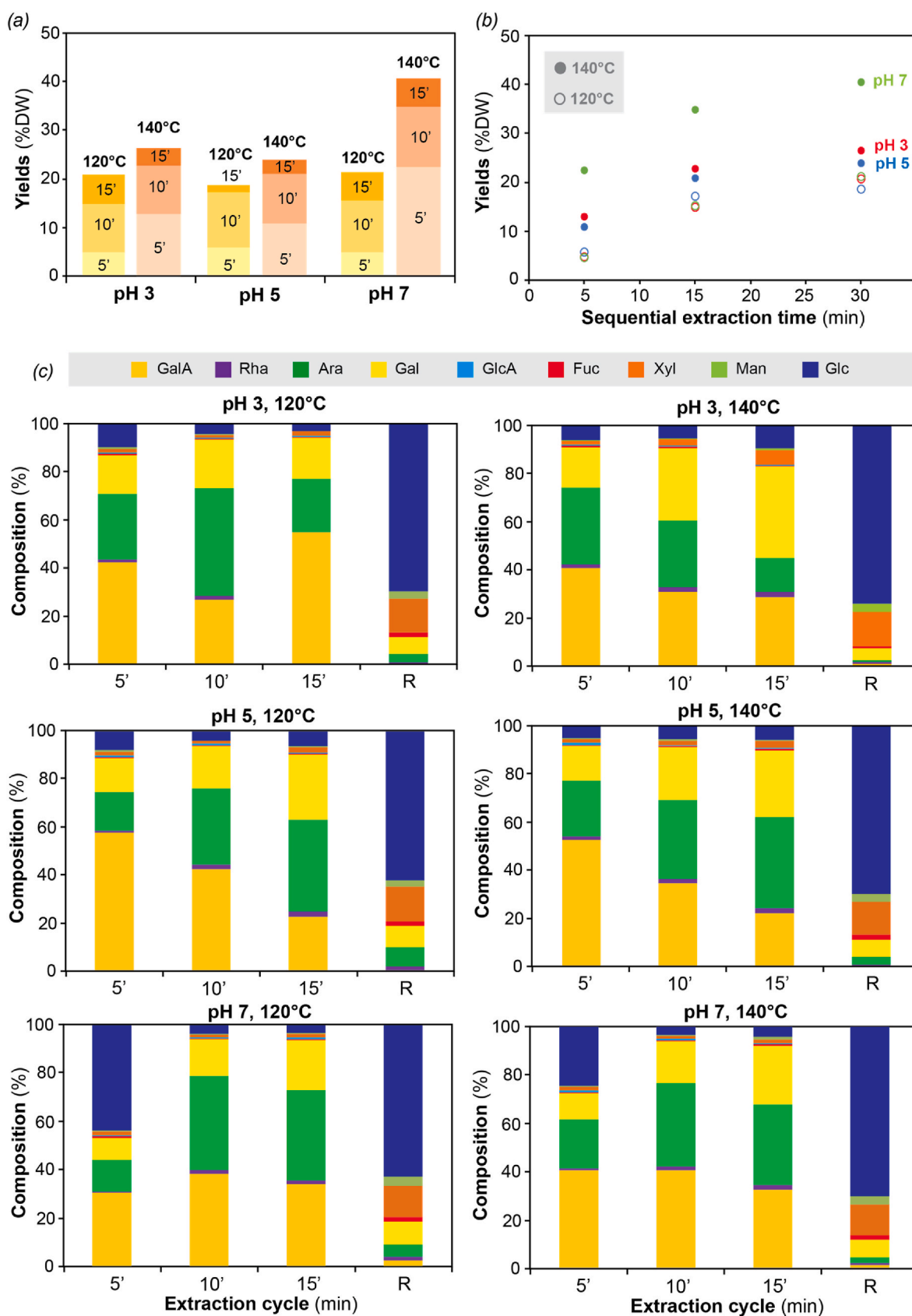


Fig. 2. Subcritical water extraction (SWE) at pH 3, pH 5 and pH 7 at 120 °C and 140 °C. (a) Overall gravimetric yields (% dry weight); (b) Time evolution of the gravimetric yields (% dry weight); (c) Monosaccharide composition (%) of extracts (5', 10' and 15') and residue (R).

Table 3

Monosaccharide composition of extracts at 120 °C and evaluation of the pectic domains.

	pH 3			pH 5			pH 7		
	5'	10'	15'	5'	10'	15'	5'	10'	15'
Fuc (mg/g)	3.8	2.6	1.9	3.5	2.5	3.7	3.0	3.5	4.7
Ara (mg/g)	149.5	325.4	127.5	85.3	208.5	298.5	80.9	265.2	281.2
Rha (mg/g)	6.3	12.8	n.d.	4.9	12.1	17.1	3.0	10.3	11.9
GlcA (mg/g)	1.5	2.5	3.0	3.5	3.0	3.2	3.8	2.6	4.9
Gal (mg/g)	88.8	147.1	96.2	73.4	114.9	213.0	57.2	103.5	153.2
Glc (mg/g)	55.3	32.1	17.8	39.8	23.4	46.4	271.9	25.7	25.3
Xyl (mg/g)	9.9	9.9	10.3	8.6	7.5	17.9	9.0	7.8	11.8
Man (mg/g)	2.8	1.7	1.0	3.3	1.9	2.8	2.2	0.7	1.6
GalA (mg/g)	233.1	194.0	310.0	303.2	274.3	173.2	188.6	261.0	255.5
Total monosaccharide (mg/g)	551.2	728.1	567.7	525.6	648.1	776.0	619.4	680.4	750.1
GalA/(Ara + Rha + Gal) ^a	0.8	0.3	1.1	1.5	0.7	0.3	1.1	0.5	0.5
Rha/GalA ^b	0.03	0.08	n.d.	0.02	0.05	0.12	0.02	0.05	0.06
(Gal + Ara)/Rha ^c	40.0	39.5	n.d.	34.4	28.4	31.7	49.3	38.3	38.8
Gal/Ara ^d	0.6	0.4	0.7	0.8	0.5	0.7	0.7	0.4	0.5

NOTE. Averages from triplicate measurements. Standard deviations presented in [Supplementary Material Table S2](#).^a Pectin linearity (Zhi, et al., 2017).^b Portion of the pectin that corresponds to the RG I domain (Zhi, et al., 2017).^c Length of the side chain linked to the RG I domain (Zhi, et al., 2017).^d Ratio of galactan and arabinan domains.

suggests that temperature can be used as a selective factor to preferentially extract pectins (at temperatures around 140 °C) and hemicelluloses (at 160 °C and above).

The presence of Glc was observed in all the time fractions with a general decreasing trend towards longer extraction times. The Glc content in the initial extracts was ascribed to easily extracted leftover starch, as previously reported (Rudjito, et al., 2019). The presence of Glc was particularly remarkable in the 5-min extracts at pH 7, which suggests that mild acidic conditions (pH 3 and 5) might either selectively extract pectic polysaccharides or induce degradation of the starch populations. Increasing the pH of the extraction to 7.0 resulted in different monosaccharide profiles than in the acidic conditions, with an increasing trend with longer extraction time. As reported earlier, in the first 5-min of SWE, more glucose was co-extracted together with HG, RG-I and RG-II. Interestingly, the Glc, Man and Xyl contents slightly increased towards the later stages of the extraction particularly at harsher acidic (pH 3) and temperature (140 °C) conditions. This suggests the increased extraction of hemicellulose fractions (e.g. xylan, xyloglucan, and mannans) in the apple pomace in these conditions (Fernandes, Silva, et al., 2019).

The time evolution of the monosaccharide composition of the

extracts at different pH and temperature conditions showed that pectin with different branching patterns could be isolated by tuning the process conditions. Different structural parameters for the pectic regions extracted at different timepoints were calculated from the results of the monosaccharide analysis, including pectin linearity, the ratio of pectin corresponding to the RG-I domain, the length of side chains linked to the RG-I domain, and the ratio of arabinan and galactan domains (Tables 3 and 4). In general, the content of GalA was higher at shorter extraction times, and progressively decreasing with longer extraction times for all pH and temperature conditions, whereas a relative overall increase of Rha, Gal, and Ara with extraction times was observed at the different conditions. This indicates that linear HG is preferentially isolated at milder conditions, whereas more complex RG-I, RG-II and potentially xyloglacturonan populations require longer extraction times. This correlates well with the general decrease in the pectin linearity factor and the overall increase in the ratio of pectin corresponding to the RG-I domain. Similar trends on the relative content of GalA and neutral sugars were observed with increasing temperature in previous subcritical water extraction of pectins from apple and citrus peel (Xin Wang, Chen, & Lü, 2014).

The high ratio of Gal and Ara with respect to Rha indicate that either

Table 4

Monosaccharide composition of extracts at 140 °C and evaluation of the pectic domains.

pH conditions	pH 3			pH 5			pH 7		
	5'	10'	15'	5'	10'	15'	5'	10'	15'
Extraction times									
Fuc (mg/g)	3.9	3.5	4.0	3.0	3.5	4.8	1.4	3.5	5.3
Ara (mg/g)	239.1	215.4	94.5	155.8	241.8	281.9	66.6	236.4	239.3
Rha (mg/g)	11.1	16.5	16.4	8.6	13.9	15.5	2.7	10.0	12.1
GlcA (mg/g)	3.3	3.1	2.6	6.8	2.4	2.8	2.4	3.3	2.3
Gal (mg/g)	125.8	234.5	254.3	98.3	161.4	206.0	35.4	116.1	174.1
Glc (mg/g)	44.5	41.2	63.5	32.3	38.5	42.1	80.1	23.8	29.0
Xyl (mg/g)	11.9	21.8	39.7	9.4	14.0	21.5	4.3	9.2	11.0
Man (mg/g)	3.6	3.3	5.3	3.4	4.5	3.5	1.5	1.5	9.3
GalA (mg/g)	302.7	239.4	191.5	356.6	253.7	164.5	131.7	272.3	230.8
Total monosaccharide (mg/g)	745.9	778.8	671.8	674.1	733.5	742.7	326.1	676.0	713.2
GalA/(Ara + Rha + Gal) ^a	0.6	0.4	0.4	1.1	0.5	0.3	1.0	0.6	0.4
Rha/GalA ^b	0.04	0.08	0.10	0.03	0.06	0.11	0.02	0.04	0.06
(Gal + Ara)/Rha ^c	34.9	28.6	22.0	31.3	30.9	33.2	39.8	37.7	36.1
Gal/Ara ^d	0.5	1.0	2.5	0.6	0.6	0.7	0.5	0.5	0.7

NOTE. Averages from triplicate measurements. Standard deviations presented in [Supplementary Material Table S3](#).^a Pectin linearity (Zhi, et al., 2017).^b Portion of the pectin that corresponds to the RG I domain (Zhi, et al., 2017).^c Length of the side chain linked to the RG I domain (Zhi, et al., 2017).^d Ratio of galactan and arabinan domains.

the branched domains of RG-I exhibit a large relative length or that free arabinogalactan populations are extracted from the apple pomace. This is in agreement with the weak and reversible bonding of arabinans and galactans to cellulose, which might then favour their extraction (Lin, Lopez-Sanchez, & Gidley, 2015). At pH 3, the Ara content shows a distinct trend for the extraction at 120 °C and 140 °C. At 120 °C, the Ara content is highest at the second cycle (10-min), with a drastic decrease at longer times, whereas at 140 °C, the Ara content decreases steadily with extraction time. This could be related to the labile nature of the arabinofuranose moieties under acidic conditions, which progressively get degraded under the subcritical water conditions. This ‘arabinose peeling’ has also been observed during subcritical water extraction of cereal arabinoxylans (Rudjito et al., 2020), resulting in arabinoxylans with lower degree of substitution. As a result, the choice of specific acidic conditions (pH 3 versus pH 5), enables to control the structure of the arabinan and galactan branches in RG-I. On one hand, the use of pH 3 promotes the extraction of RG-I with more galactan content, whereas the use of moderate pH conditions (pH 5) enable the enrichment of the arabinan segments. This is clearly observed in the ratio of galactan and arabinan domains in the hairy regions (Tables 3 and 4).

All in one, these results indicate that controlled subcritical water treatment of apple pomace results in higher yields and purities of extracted pectin compared to traditional hot acidic extraction. Indeed, previous studies using different hot acidic solutions (between pH 1 and pH 3, at temperatures between 65 °C and 100 °C) result in the extraction of pectins with yields between 3 and 14%, requiring in most cases much longer extraction times (between 1 h and 4 h (Perussello, et al., 2017)). Moreover, controlling the time conditions during SWE enable the

fractionation of pectins with different branching structures, with HG being extracted at shorter times and RG-I being enriched at longer extraction times. Finally, the choice of acidic conditions also tunes the relative ratio of arabinan and galactan branches in RG-I, since arabinan is more labile and susceptible to undergo degradation at more acidic and higher temperature conditions.

3.3.2. Time evolution of the molar mass distributions during SWE

The molar mass distributions, the number- (M_n) and weight-average (M_w) molar masses of the extracted fractions are presented in Fig. 3 and Supplementary Table S1, respectively. Complex multimodal size distributions and apparent molar masses ranging from 10^3 Da to 10^6 Da were observed for almost all temperature and pH conditions, indicating a complex range of macromolecular populations. Fruit pectins usually exhibit heterogeneous molar mass distributions between 10,000 and 700,000 Da depending on the fruit source and extraction technique (Colodel & Petkowicz, 2019; Wikiera, Mika, Starzynska-Janiszewska, & Stodolak, 2015), which is similar to the values here reported for the extracted fractions at 120 °C and 140 °C.

In general, it is extremely difficult to assign specific macromolecular populations in terms of molar mass/size to their polysaccharide origin, but general trends can be observed from the apparent molar mass distributions from SEC. Considering the extracts at pH 7, a larger peak was observed in the high molar mass region (between 10^5 – 10^6 Da) of the 5-min extract as compared to the pH 3.0 and pH 7.0 extracts, which disappeared with prolonged extraction time (Fig. 3a). This is in alignment with the high glucose content of the 5-min extract at pH 7, corresponding to starch and β -glucans (Fig. 2c and Table 3). At longer

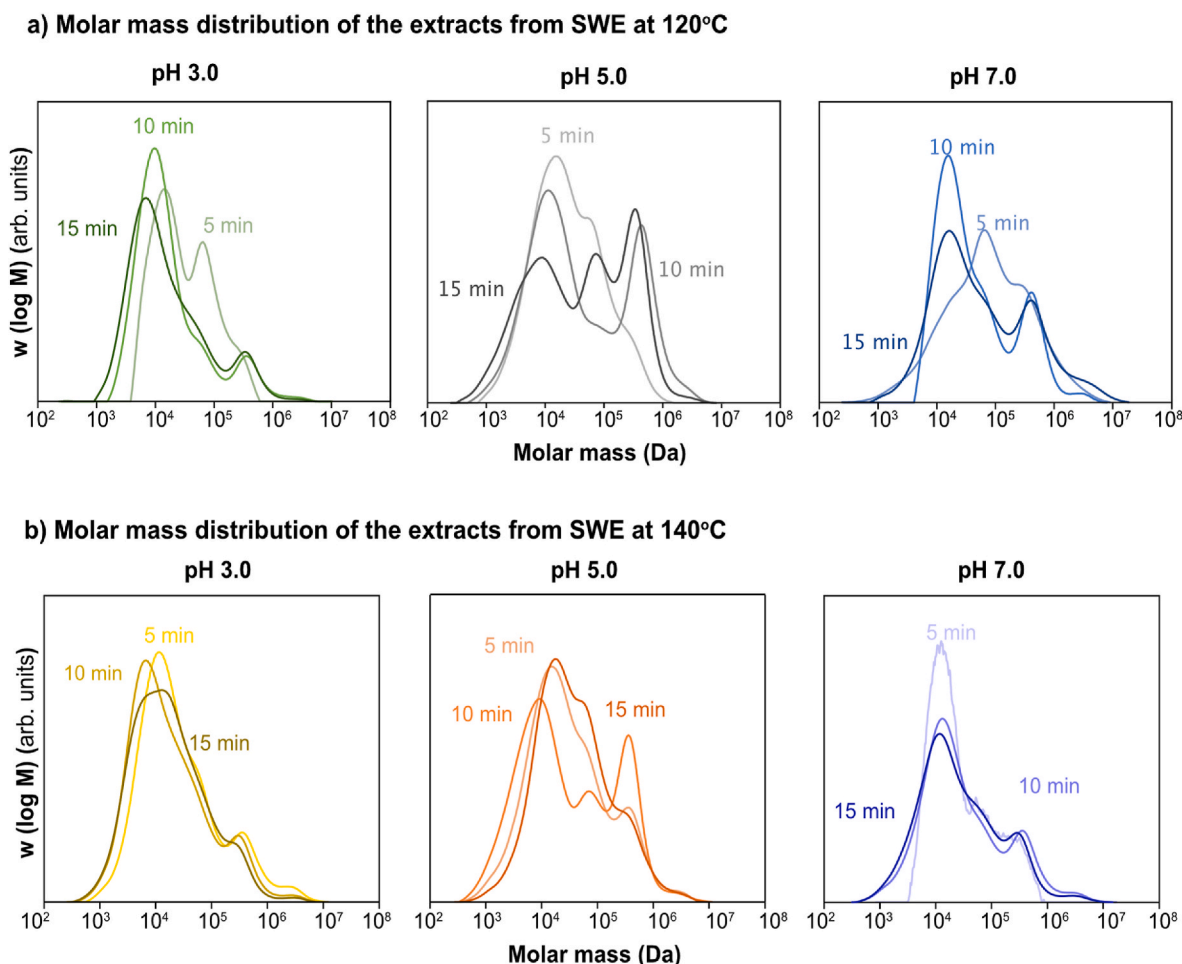


Fig. 3. Molar mass distributions of SW extracts at pH 3, pH 5 and pH 7: (a) 120 °C and (b) 140 °C.

extraction times under neutral conditions, the molar mass distributions did not significantly change as the extraction progressed, which indicate the limited occurrence of hydrolysis as a result of the mild extraction conditions.

At pH 5.0, the time evolution of the molar mass distributions of the extracts changed from bimodal to trimodal with increasing extraction time, which confirmed the presence of various pectic regions. Interestingly, the peak corresponding to larger macromolecular populations (between 10^5 to 10^6 Da) increased at longer extraction times especially at 120 °C, which might be assigned to an enrichment in the branched RG-I domains. Indeed, in a previous work, we performed preparative fractionation of extracted pectins from spruce bark in terms of their macromolecular, and we performed detailed molecular characterization of the different pectic fractions (Le Normand, Rietzler, Vilaplana, & Ek, 2021). The larger molar mass populations could be ascribed to highly

branched RG-I domains, whereas the fractions with lower molar masses were enriched with linear homogalacturonan domains and free arabinogalactan populations.

The extracts at acidic conditions (pH 3) exhibit multimodal distributions as well, although with clear differences compared to the other extraction conditions. The smaller macromolecular populations (between 10^3 and 10^4) are clearly enriched with increasing extraction time, suggesting that hydrolysis of the polysaccharide chains was promoted in these acidic extractive conditions (Colodel et al., 2019). This trend is clearly observed at higher extractive temperatures, in agreement with the hydrolytic effect of increasing temperature on molar mass previously observed for other biomass sources (Rudjito et al., 2020). In general, neutral (pH 7) or moderate acidic conditions (pH 5) under subcritical water are able to preserve the molar mass of the isolated polysaccharide populations, whereas more aggressive acidic conditions result in higher

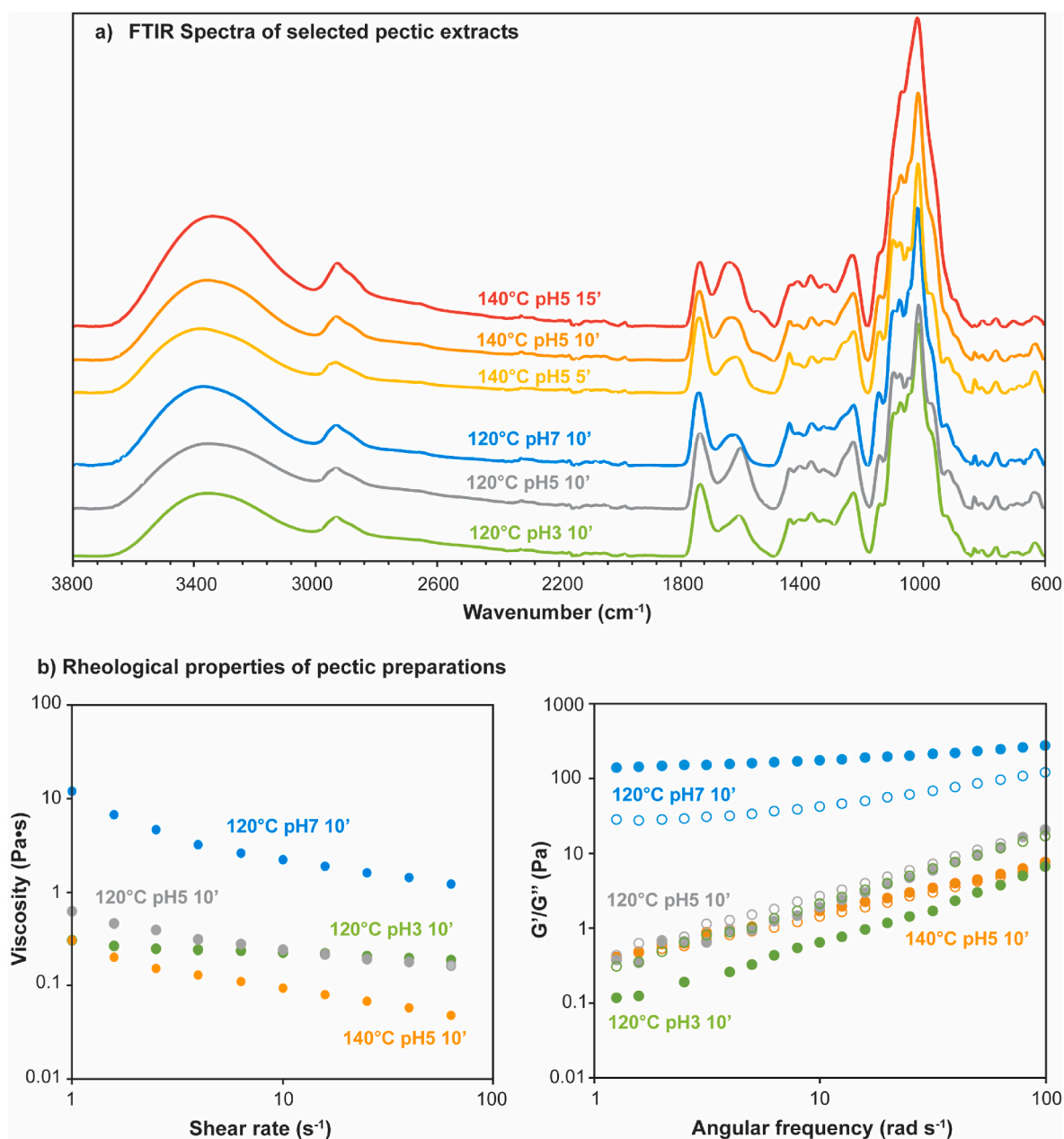


Fig. 4. a) Chemical structure of selected subcritical water extracts and (b) Viscosity, storage modulus (G') (solid symbols) and loss modulus (G'') (open symbols) of the samples prepared from the selected extracts at pH 3.5 and with 60% sucrose. NOTE: the pectin sample extracted at 120 °C pH3 10' exhibited a torque irregularity at the 3rd and 5th measurements, so such datapoints have been removed.

extraction yields at the expense of hydrolytic degradation.

3.3.3. Chemical structure and degree of methylation of selected pectic extracts

Based on the purity of the pectic components and the results from the molar mass distributions, the 10-min extracts of SWE at 120 °C and all the time extracts of SWE at pH 5 at 140 °C were selected to be analyzed in terms of their chemical structure by FTIR (Fig. 4a). FTIR analysis showed typical spectral bands for polysaccharides between 3600 and 3000 cm^{-1} , and between 3000 and 2800 cm^{-1} corresponding to the OH-stretching and C-H stretching, respectively. The C-H stretching vibrations superimposed upon the broader O-H bands, which was in agreement with Gnanasambandam et al. (Gnanasambandam & Proctor, 2000). The peaks seen in all the extracted fractions at 2950–2750 cm^{-1} , 1720 cm^{-1} and 1640–1520 cm^{-1} were attributed to O-CH₃, ester carbonyl (C=O) and carboxylate ion stretching (COO⁻), respectively, as the extracted pectins were methyl esterified (Gnanasambandam et al., 2000). The peaks between 1300 and 700 cm^{-1} represented the fingerprint region of pectin, corresponding to glycosidic linkages, C-O-C vibration, C-C and C-O stretching.

The 10-min extracts of SWE at 120 °C and of SWE at pH 5 at 140 °C were characterized in terms of their degree of methylation (DM) (Table 5). The pH 3.0 and pH 5.0 extracts showed lower DM (<50%), corresponding to low-methoxyl pectin, whereas the 10-min extract of pH 7.0 had the highest DM corresponding to a high-methoxyl pectin. The lower DM of the extract at 120 °C likely resulted from acid hydrolysis of pectins as discussed earlier. The pH conditions appeared to have a substantial effect on the DM of the extracted pectins, i.e. acidic pH possibly resulted in the cleavage of methyl groups from the galacturonan chains (Sahari, Akbarian, A., & Hamed, 2003). On the other hand, higher extraction temperature resulted in somewhat higher DM in agreement with Wang et al. X. Wang, Chen, and Lü (2014), which could be attributed to the higher efficiency of subcritical water at higher temperatures resulting in the recovery of more recalcitrant methylated pectin populations (Fig. 2b and c). These four extracts were considered representative of the whole extracts at different pH and temperature conditions in SWE and used for the determination of their gelling behavior.

3.4. Gelling behavior of apple pomace pectins extracted by subcritical water

The rheological properties of the samples prepared from the selected extracts are presented in Fig. 4b. All samples showed shear thinning behavior, which can be attributed to the weak associations among pectin molecules that can easily be disrupted, resulting in a decreased viscosity with increasing shear rate (Gan, Abdul Manaf, & Latiff, 2010). Among all the samples, 120 °C-pH 7–10 min showed the highest viscosity, which arose from its high DM (Table 3).

The storage modulus (G') and elastic modulus (G'') of the samples prepared from the 10-min extracts were also influenced by the DM. As expected, the sample prepared from 120 °C-pH 3–10 min exhibited typical behavior of a solution ($G' < G''$), indicating the restricted inter-chain entanglement of pectin molecules due to their low DM. Contrarily, the extract at 120 °C-pH 7–10 min showed an apparent gel behavior in

the presence of sucrose at pH 3.5 ($G' > G''$), which could be again related to the high DM of the sample. The samples 120 °C-pH 5–10 min and 140 °C-pH 5–10 min, on the other hand, showed overlapping G' and G'' curves over the frequency range ($G' = G''$), with a similar slope for both moduli. This suggested that these samples passed through the point of critical crosslinking (Gilsenan, Richardson, & Morris, 2000). The effect of pH on the rheological properties is less significant than the influence of the DM. At pH 3.5, the carboxyl groups of these two extracts could be partially converted to the protonated form, which might suppress electrostatic repulsion and facilitate intermolecular interactions (Gilsenan, et al., 2000; Lootens et al., 2003). However, the degree of this protonation does not seem to be sufficient to induce complete gelation of these extracts.

3.5. Emulsifying properties of apple pomace pectins extracted by subcritical water

The emulsifying properties and long term stability of two selected apple pomace pectin fractions extracted using subcritical water, with high methoxyl content (extract 120 °C-pH 7–10 min with DM of 89.1%) and low methoxyl content (extract 140 °C-pH 5–10 min with DM of 28.4%) were evaluated by optical microscopy and laser diffraction at different storage days. The results were compared and benchmarked with a commercial high methoxyl citrus pectin (DM of 71%). In general, the emulsifying properties of pectin fractions depends on the molecular structure of the polysaccharide in terms of molar mass, relative content of linear HG and branched RG-I domains, the size and conformation of the RG-I arabinogalactan side chains, the content of methyl, acetyl and feruloyl substitutions, and the presence of covalently-bound protein domains (Kpodo, et al., 2018; Ngoumazong, Christiaens, Shpigelman, Van Loey, & Hendrickx, 2015). Indeed, a balance of the hydrophilic and hydrophobic moieties in the pectic fractions is expected to stabilize the oil and water interfaces, contributing to improved emulsifying ability and stability.

The emulsions from the commercial citrus pectin (Citrus HMP) showed excellent homogeneity with uniform droplet formation and unimodal particle size distributions of 1 μm (Fig. 5). The control emulsions were very stable during storage conditions, and the particle size distribution at the different storage days were almost superimposed. The apple pomace fraction with high methoxyl content (Apple HMP) also showed homogeneous emulsions with uniform droplet size of approximately 1 μm . This could be attributed to the high DM content of this apple pomace extract, which contributes to the formation of fine droplets stabilized by both the hydrophilic and hydrophobic components of the pectin. However, already after 1 day the stability of the emulsions decreased and the formation of larger particles of 10 μm by coalescence was observed. The stability of such emulsions was however conserved after longer storage time, which indicate the potential of these high methoxylated apple pomace pectins as alternative emulsifying agents. The lower stability of the apple pomace pectin fractions compared to the commercial citrus pectin may be due to the purity and the presence of multiple polymeric populations from the molar mass distributions, as the SWE fraction still contained minor amounts of starch and other polysaccharide impurities (Table 3). Indeed, lower purity and the presence of starch has been reported to hinder the emulsifying properties of pectic extracts from hawthorn berry (Roman, et al., 2021). Moreover, the balance between linear HG and branched RG-I domains also contributes to the emulsifying stability of the pectins, as the RG-I domains provide steric stabilization, whereas the HG domains contribute to electrostatic stabilization (Ngoumazong, et al., 2015). Indeed, a HG/RG-I ratio above 2 has been reported to improve the long term stability of emulsions (Kpodo, et al., 2018). In the case of the Apple HMP, it seems that the higher relative abundance of RG-I domains with higher flexibility compared to the linear HG domains seem to result in shorter separation distances between droplets, which might facilitate coalescence during long term storage. Finally, the apple pomace extract

Table 5
Degree of methylation (DM) of selected extracts.

Sample	DM (%)
120 °C-pH 3–10 min	8.1 ± 0.1
120 °C-pH 5–10 min	19.5 ± 0.4
120 °C-pH 7–10 min	89.1 ± 0.3
140 °C-pH 5–10 min	28.4 ± 0.3

NOTE. The results are presented as averages ± standard deviation.

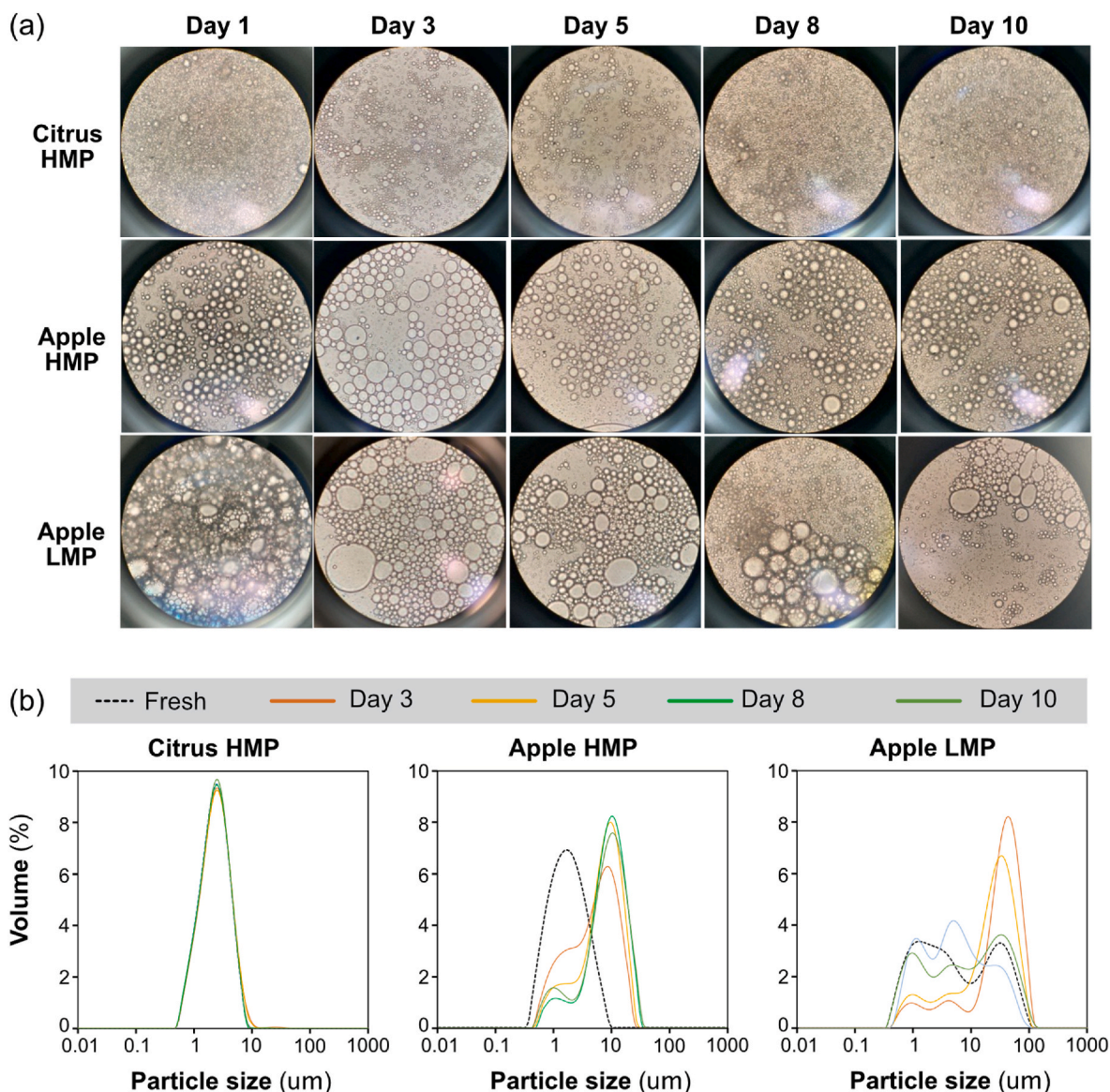


Fig. 5. Emulsifying properties of two pectin fractions from apple pomace with high methoxyl content (Apple HMP, 120°C-pH 7–10 min) and low methoxyl content (Apple LMP, 120°C-pH 3–10 min) extracted by subcritical water compared to a commercial high methoxyl pectin from citrus (Citrus HMP). (a) Optical microscopy of the oil-in-water emulsions at different storage days (b) Droplet size distribution of pectin emulsions fresh and at different storage days (3, 5, 8, 10 days).

with low methoxyl content (Apple LMP) exhibited the worst emulsifying properties and long-term stability, as observed by the heterogeneous and multimodal size distribution of the droplets with multiple particle size populations. Indeed, the emulsions seemed to increasingly coalesce into large particles of sizes between 10 and 100 μm . Moreover, the emulsions from Apple LMP showed creaming behaviour (Supplementary Material Fig. S3), evidencing the poor emulsifying properties of this fraction. This could be again attributed to the heterogeneous composition of this pectic extract, the low methoxyl content, and the low relative content of linear HG domains.

4. Conclusions

This study has shown an efficient way of the valorization of a food waste, i.e. apple pomace, into natural antioxidants and pectin as valuable sources for novel bioactive and techno-functional materials. The extraction of polyphenols from apple pomace was demonstrated to be simple and fast only using 50% ethanol. Chlorogenic acid was identified as the main phenolic acid compound with a significant antioxidant

activity. Subcritical water extraction (SWE) was successfully implemented for the extraction of pectic polysaccharides at different pH, time and temperature conditions. The different conditions in SWE produced different pectin populations with different molecular structure in terms of presence of linear HG and branched RG-I domains, content of methyl group substitutions and molecular size. Linear HG was preferentially isolated at milder conditions, whereas more branched RG-I populations require longer extraction times. Acidic extraction generated low methoxyl pectin that can be potentially used as a gelling agent in low-sugar/low-calorie food products. Neutral pH in SWE produced high methoxyl pectin when combined with high extraction temperature, which yielded the formation of a pectin gel in the presence of sugar at low pH. The emulsifying properties of the apple pomace pectins with high and low methoxyl content were evaluated. The apple pomace pectin with higher methoxyl content showed higher emulsifying ability compared to the low methoxyl extract, resulting in a homogeneous particle size distribution of the droplets upon formation. This study shows that adjusting the time, pH and temperature conditions during subcritical water extraction has large potential to fine tune the

molecular structure of pectic polysaccharides in terms of linear HG and branched RG-domains, the ratio of arabinan and galactan side chains, and the methoxyl content, which has a large influence on the techno-functional properties of these food hydrocolloids as gelling and emulsifying agents.

CRediT author statement

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Francisco Vilaplana reports financial support was provided by Swedish Research Council Formas. Francisco Vilaplana reports a relationship with Oatly AB that includes: employment.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foodhyd.2023.109148>.

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